



Modelling Financial Markets

Chapter 11: Continuous-Time Models



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARFIMA: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BSS: Brownian SemiStationary (process)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CBOE: Chicago Board Options Exchange
- ▣ CI: Confidence Interval
- ▣ CIR: Cox–Ingersoll–Ross (model)
- ▣ CKLS: Chan–Karolyi–Longstaff–Sanders (model)
- ▣ CLT: Central Limit Theorem
- ▣ DGS10: 10-Year Treasury Constant Maturity Rate (FRED series code)
- ▣ DTB3: 3-Month Treasury Bill: secondary market rate (FRED series code)
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FRED: Federal Reserve Economic Data
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GBM: Geometric Brownian Motion
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ JB: Jarque–Bera (test)
- ▣ LLM: Large Language Model
- ▣ LR: Likelihood Ratio (test)
- ▣ MCS: Model Confidence Set
- ▣ MLE: Maximum Likelihood Estimation
- ▣ MSE: Mean Squared Error

Acronyms used in this material (2/2)

- ▣ **NLLS: Non-Linear Least Squares**
- ▣ **OLS: Ordinary Least Squares**
- ▣ **OU: Ornstein–Uhlenbeck (process)**
- ▣ **PDE: Partial Differential Equation**
- ▣ **QLIKE: Quasi-LIKelihood (loss function)**
- ▣ **QQ: Quantile–Quantile (plot)**
- ▣ **RFSV: Rough Fractional Stochastic Volatility (model)**
- ▣ **SDE: Stochastic Differential Equation**
- ▣ **SVJ: Stochastic Volatility with Jumps (model)**
- ▣ **US: United States**
- ▣ **VIX: CBOE Volatility Index**
- ▣ **XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)**

Chapter Route and Outcomes

The question and the route

- ▣ Prices are recorded daily or every few milliseconds: can one model describe them at **every** time scale at once?
- ▣ Part I: Brownian motion, Itô calculus, simulation, geometric Brownian motion, mean reversion
- ▣ Part II: estimating diffusions, change of measure, jumps, stochastic volatility, models against data, a case study, AI-assisted discovery
- ▣ History of the field and technical detail: [▶ Appendix](#), before the references

After this chapter you can

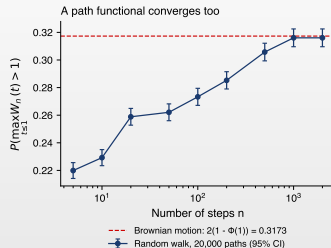
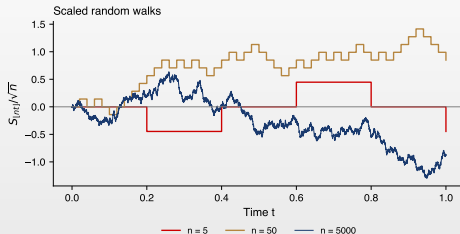
- ▣ apply Itô's lemma and simulate a stochastic differential equation with a known error order
- ▣ estimate GBM, Ornstein–Uhlenbeck, Merton and Heston on data, with standard errors that respect what the sample identifies
- ▣ say which feature of the data each model reproduces, and which feature none of them does
- ▣ Option pricing ([Chapter 12](#)), interest-rate models and risk management are written in this language

Data Used in This Chapter

Series	Type	Period	Days
S&P 500	US equity index	1990–2026	9 245
BET	BVB equity index	2000–2026	6 670
Bitcoin	crypto-asset, 7 days a week	2014–2026	4 384
VIX	CBOE implied volatility index of the S&P 500	1990–2026	daily
DTB3	3-month US Treasury bill rate (FRED)	1954–2026	daily

- Daily closing values up to 18 September 2026; daily log returns $r_t = \ln P_t - \ln P_{t-1}$, each series on its own calendar (cleaning rules: [Appendix](#))
- Time in years, daily step $\Delta t = 1/252$ (Bitcoin: $1/365$); drifts and volatilities annualised
- W_t : standard Brownian motion; S_t : price; v_t : instantaneous variance; $\mathcal{F}_t$: the information (filtration) up to t

From Random Walk to Brownian Motion: Donsker's Theorem



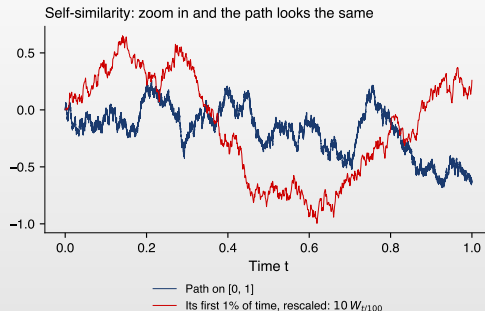
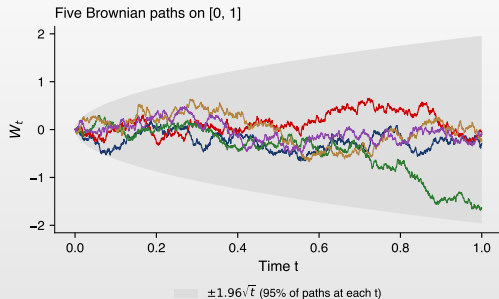
- Scaled random walk $W_n(t) = S_{[nt]}/\sqrt{n}$, $S_k = \xi_1 + \dots + \xi_k$, steps i.i.d. with mean 0 and variance 1: time step $1/n$, space step $1/\sqrt{n}$
 - Central limit theorem (CLT): $W_n(t) \rightarrow N(0, t)$ for each t ; Donsker: $W_n \Rightarrow W$ for the **whole path**, whatever the law of the steps Billingsley (1999)
- Right panel: $P(\max_{t \leq 1} W_n(t) > 1)$ rises from 0.220 ($n = 5$) to 0.316 ($n = 2000$), towards the reflection-principle value $2(1 - \Phi(1)) = 0.3173$ (Φ : standard Normal distribution function)
- A coarse grid misses the maximum between observations and convergence is slow, of order $1/\sqrt{n}$: daily closes understate the intraday range



Brownian Motion: Definition and Properties

- **Standard Brownian motion:** $W_0 = 0$; independent increments; $W_t - W_s \sim N(0, t - s)$ for $s < t$; continuous paths
 - ▶ Existence: [Wiener \(1923\)](#); simulation on a grid: $W_{t+\Delta t} = W_t + \sqrt{\Delta t} Z$, $Z \sim N(0, 1)$
- $\text{Cov}(W_s, W_t) = \min(s, t)$; a Gaussian process and a **martingale**: $\mathbb{E}[W_t | \mathcal{F}_s] = W_s$ (the efficient-market random walk of [Chapter 2](#))
- **Self-similarity:** $\{c^{-1/2} W_{ct}\}$ is again a standard Brownian motion for any $c > 0$
- Paths are continuous but **nowhere differentiable**
 - ▶ $(W_{t+h} - W_t)/h$ has standard deviation $1/\sqrt{h} \rightarrow \infty$: ordinary calculus fails

Brownian Paths



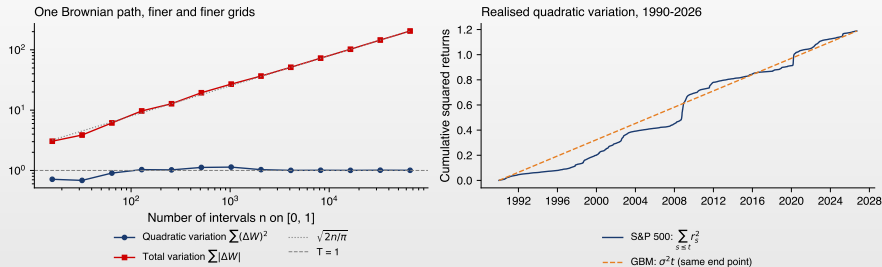
- Left: five paths with the pointwise band $\pm 1.96\sqrt{t}$ (95% for each t separately, not for the whole path); right: a path and its first 1% of time, stretched 100 times in time and 10 times in space
- The band widens like $\sqrt{t}$ (the square-root-of-time rule of [Chapter 7](#)) and a zoomed piece looks like the whole path; real markets break self-similarity, since fat tails and clustering depend on the horizon ([Chapter 1](#))



Total and Quadratic Variation

- Grid $0 = t_0 < t_1 < \dots < t_n = T$ with step $\Delta t = T/n$ and increments $\Delta W_k = W_{t_k} - W_{t_{k-1}}$
- **Total variation** $\sum_k |\Delta W_k|$: the length of the path
 - ▶ Its mean is $\sqrt{2nT/\pi} \rightarrow \infty$: a Brownian path has infinite length
- **Quadratic variation** $[W]_T = \lim \sum_k (\Delta W_k)^2 = T$
 - ▶ A random sum with a **deterministic** limit
 - ▶ $\text{Var}(Q_n) = 2T^2/n$: one year of daily data gives about 9% relative error ($\sqrt{2/252}$), the idea behind realised variance ([Chapter 9](#))
- Shorthand used in Itô calculus: $(dW_t)^2 = dt$

Quadratic Variation: Simulation and Data



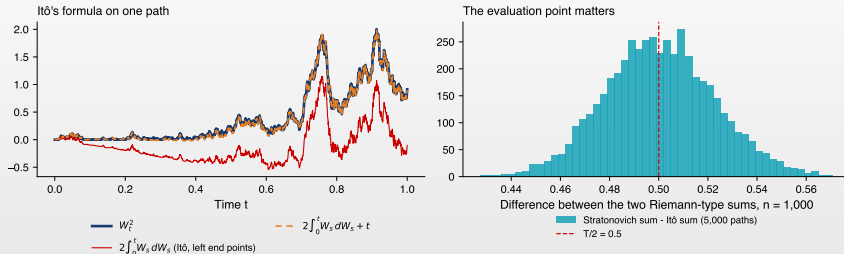
- Left: one Brownian path on grids with 16 to 65,536 intervals: the quadratic variation settles at 1.006 ($T = 1$), while the total variation keeps growing, 205 against $\sqrt{2n/\pi} = 204$
- Right: cumulative $\sum r_t^2$ of the S&P 500, 1.188 over 36.7 years (average volatility 18.0%); under geometric Brownian motion it would follow the straight line $\sigma^2 t$
- It rises in steps: September 2008 – March 2009 is 1.6% of the days but 15.2% of the quadratic variation; February – April 2020: 0.5% and 7.8%. Volatility is itself random (Heston, Section 10)

The Itô Integral

- Goal: give a meaning to $\int_0^T H_t dW_t$, the gain of holding H_t units of an asset whose price moves like W
 - ▶ Riemann–Stieltjes integration fails because the path has infinite total variation
- **Itô integral**: the limit in mean square of sums with the integrand at the **left** end point
 - ▶ $\int_0^T H_t dW_t = \lim_{n \rightarrow \infty} \sum_{k=1}^n H_{t_{k-1}} (W_{t_k} - W_{t_{k-1}})$, for H **adapted** with $\mathbb{E} \left[\int_0^T H_t^2 dt \right] < \infty$
 - ▶ Financial reading: H_t uses only information up to t ; the position is chosen before the next price move
- Zero mean, $\mathbb{E} \left[\int_0^T H_t dW_t \right] = 0$, and the **Itô isometry**, $\mathbb{E} \left[\left(\int_0^T H_t dW_t \right)^2 \right] = \mathbb{E} \left[\int_0^T H_t^2 dt \right]$
 - ▶ $t \mapsto \int_0^t H_s dW_s$ is a martingale: no strategy earns money from a martingale on average (weaker conditions:

▶ Appendix
- Mid-point sums give the **Stratonovich** integral, a different number; finance uses Itô, because a mid-point position would look into the future

Worked Example: $\int_0^T W_t dW_t$ on Simulated Paths



- Each term: $W_{t_{k-1}} \Delta W_k = \frac{1}{2}(W_{t_k}^2 - W_{t_{k-1}}^2) - \frac{1}{2}(\Delta W_k)^2$; the first part telescopes, the second tends to $\frac{1}{2}T$:

$$\int_0^T W_t dW_t = \frac{1}{2}W_T^2 - \frac{1}{2}T$$
 - Ordinary calculus would give $\frac{1}{2}W_T^2$: the extra $-\frac{1}{2}T$ is the quadratic variation, and the mean is $\frac{1}{2}(T - T) = 0$
- Left: on one path, $2 \int_0^t W dW + t$ lies exactly on W_t^2 ; right: Stratonovich minus Itô sums over 5,000 paths, mean 0.500, s.d. 0.022 (theory $T/2 = 0.5$)
- Itô sums: mean 0.007, variance 0.520; theory: 0 and, by the isometry, $\int_0^1 t dt = 1/2$

Itô Processes and Itô's Lemma

- **Itô process:** $dX_t = a_t dt + b_t dW_t$ (drift a_t , diffusion coefficient b_t , both adapted)
 - ▶ Shorthand for $X_t = X_0 + \int_0^t a_s ds + \int_0^t b_s dW_s$
- **Itô's lemma:** for $f(t, x)$ twice differentiable [Itô \(1944\)](#)
 - ▶ $df(t, X_t) = (f_t + a_t f_x + \frac{1}{2} b_t^2 f_{xx}) dt + b_t f_x dW_t$
- Rule of thumb: Taylor expansion to second order with $dt \cdot dt = 0$, $dt \cdot dW = 0$, $dW \cdot dW = dt$
 - ▶ The second-order term $\frac{1}{2} b_t^2 f_{xx} dt$ does not vanish: this is the whole novelty

Quadratic Covariation and the Multivariate Itô Formula

- **Quadratic covariation** $[X, Y]_t = \lim \sum_k \Delta X_k \Delta Y_k$; if $\text{Corr}(dW^{(1)}, dW^{(2)}) = \rho$, then $d[W^{(1)}, W^{(2)}]_t = \rho dt$
 - ▶ Product rule: $d(X_t Y_t) = X_t dY_t + Y_t dX_t + d[X, Y]_t$
- **Multivariate Itô formula** for $X_t \in \mathbb{R}^d$ and $f(t, x)$ twice differentiable
 - ▶ $df(t, X_t) = f_t dt + \sum_i f_{x_i} dX_t^i + \frac{1}{2} \sum_{i,j} f_{x_i x_j} d[X^i, X^j]_t$
- Worked example (Heston, Section 10): $dS = \mu S dt + \sqrt{v} S dW^{(1)}$, $dv = \kappa(\theta - v) dt + \xi \sqrt{v} dW^{(2)}$
 - ▶ $d(S_t v_t) = S_t dv_t + v_t dS_t + \rho \xi v_t S_t dt$
 - ▶ ρ enters the drift of the product: with $\rho < 0$, price and variance tend to move in opposite directions

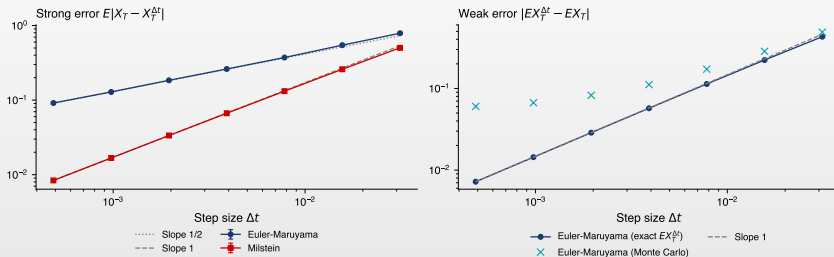
Stochastic Differential Equations and Two Schemes

- A **stochastic differential equation** (SDE): $dX_t = a(t, X_t) dt + b(t, X_t) dW_t$, X_0 given: drift and diffusion depend on the current state
 - ▶ Exact transitions for GBM, Ornstein–Uhlenbeck and Merton: simulate them exactly, with no discretisation error; Heston has only a closed-form characteristic function; existence conditions: [Appendix](#)
- **Euler–Maruyama**: freeze a and b over each step [Higham \(2001\)](#)
 - ▶ $X_{k+1} = X_k + a(X_k) \Delta t + b(X_k) \Delta W_k$, $\Delta W_k = \sqrt{\Delta t} Z_k$
- **Milstein**: add the next Itô–Taylor term [Milstein \(1975\)](#)
 - ▶ $X_{k+1} = X_k + a(X_k) \Delta t + b(X_k) \Delta W_k + \frac{1}{2} b(X_k) b'(X_k) [(\Delta W_k)^2 - \Delta t]$
 - ▶ The correction has conditional mean zero: it raises the strong order (path accuracy), not the weak order

Strong and Weak Convergence

- **Strong error:** $\mathbb{E}|X_T - X_T^{\Delta t}| \leq C \Delta t^\gamma$, how close each simulated path is to the true one
 - ▶ Euler–Maruyama $\gamma = 1/2$; Milstein $\gamma = 1$; needed for path-dependent quantities: hedging errors, barrier crossings
- **Weak error:** $|\mathbb{E}[g(X_T^{\Delta t})] - \mathbb{E}[g(X_T)]| \leq C \Delta t^\beta$, how close the distribution is
 - ▶ Both schemes: $\beta = 1$; enough for prices of European options and other expected values
 - ▶ Conditions: smooth coefficients with bounded derivatives and a smooth g Kloeden & Platen (1992); square-root coefficients (CIR, Heston), quantiles such as VaR and barrier crossings can converge more slowly
- Test equation Higham (2001): $dX = \lambda X dt + \mu X dW$, $\lambda = 2$, $\mu = 1$, $X_0 = 1$, $T = 1$, with the exact solution as benchmark
 - ▶ On a log-log plot, the order is the slope of the error against the step size (next slide)

Convergence of the Schemes

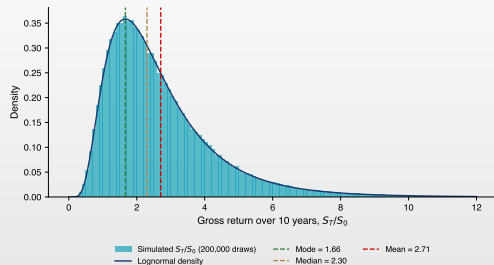
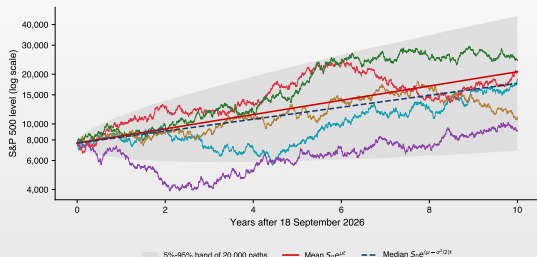


- 20,000 Brownian paths on a fine grid of 2^{11} steps, coarsened to steps from 2^{-11} to 2^{-5} ; strong orders (slope of the log-log fit, 95% CI by resampling whole paths): Euler–Maruyama 0.518 [0.512, 0.525], Milstein 0.986 [0.984, 0.987]
 - ▶ The narrow CIs measure Monte Carlo noise only; the small gap to 1/2 and 1 is finite-step curvature, not a different order; halving Δt halves the Milstein error, but cuts the Euler error only by 29%
- Weak order of Euler–Maruyama from the exact mean $(1 + \lambda \Delta t)^{T/\Delta t}$: 0.985, a deterministic slope
 - ▶ The Monte Carlo estimates flatten at small steps: their standard error, about 0.068, exceeds the bias; the sample size, not the step, is often the binding constraint [Glasserman \(2003\)](#)

Geometric Brownian Motion (GBM) from Itô's Lemma

- SDE: $dS_t = \mu S_t dt + \sigma S_t dW_t$: the **percentage** change has constant drift μ and volatility σ per year
 - ▶ Proposed by [Osborne \(1959\)](#) and [Samuelson \(1965\)](#); the price model behind [Black & Scholes \(1973\)](#)
- Itô's lemma for $f(x) = \ln x$, with $f_x = 1/x$, $f_{xx} = -1/x^2$, $a_t = \mu S_t$, $b_t = \sigma S_t$
 - ▶ $d \ln S_t = (\mu - \frac{1}{2}\sigma^2) dt + \sigma dW_t$, so $S_t = S_0 \exp((\mu - \frac{1}{2}\sigma^2)t + \sigma W_t)$
 - ▶ Prices stay positive; exact simulation needs only Normal draws
- The log price grows at the log drift $m = \mu - \sigma^2/2$, not at μ : volatility lowers the growth of the median investor's wealth
 - ▶ $\sigma = 18\%$ costs $\sigma^2/2 = 1.6$ percentage points a year; $\sigma = 67\%$ (Bitcoin), 22 points
- Testable implications: $r_t = \ln(S_t/S_{t-\Delta t})$ i.i.d. $N(m\Delta t, \sigma^2\Delta t)$ at every frequency
 - ▶ Skewness 0, excess kurtosis 0; no autocorrelation in r_t , $|r_t|$ or r_t^2 ; variance proportional to the horizon

Lognormal Outcomes: Ten Years of Simulated S&P 500 Paths



- GBM with $\hat{\mu} = 10.0\%$, $\hat{\sigma} = 18.0\%$ (1990–2026), $T = 10$ years: mean $\mathbb{E}[S_T] = S_0 e^{\mu T}$, median $S_0 e^{mT}$, mode $S_0 e^{(m-\sigma^2)T}$; gross returns 2.71, 2.30, 1.66 (right)
- Left: 20,000 exact paths from $S_0 = 7\,650$ (18 September 2026): mean 20 696, median 17 604, 5%–95% range 6 878 – 45 017; $P(S_T < S_0) = \Phi(-m\sqrt{T}/\sigma) = 7.1\%$
- $P(S_T < \mathbb{E}[S_T]) = \Phi(\sigma\sqrt{T}/2) = 61.2\%$: most investors end below the average, so report the median and quantiles; the fan is too narrow in crises (no fat tails, no clustering: tested below)

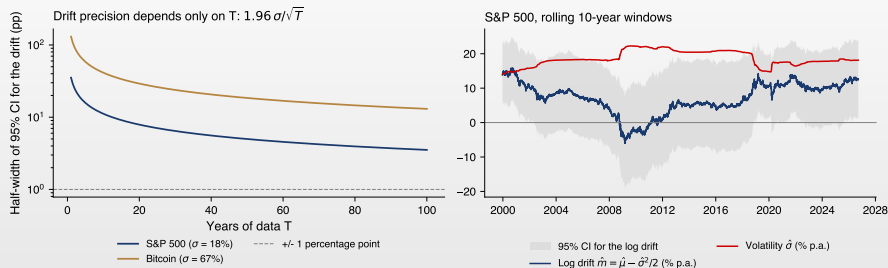
Estimating GBM by Maximum Likelihood

- Maximum likelihood estimation (MLE) from n log returns at step Δt , over $T = n\Delta t$ years: $\hat{\sigma}^2 = \frac{1}{n\Delta t} \sum_t (r_t - \bar{r})^2$, $\hat{m} = \bar{r}/\Delta t$, $\hat{\mu} = \hat{m} + \hat{\sigma}^2/2$
- Under GBM, the standard error (s.e.) $\text{s.e.}(\hat{\sigma}) = \sigma/\sqrt{2n}$ improves with more observations; $\text{s.e.}(\hat{m}) = \sigma/\sqrt{T}$ depends only on the **calendar span**, since $\sum_t r_t = \ln(S_T/S_0)$ uses only the first and the last price

	Years	$\hat{\sigma}$ (%)	$\text{s.e.}(\hat{\sigma})$	$\hat{m}$ (%)	$\text{s.e.}(\hat{m})$
S&P 500	36.7	18.0	0.13	8.3	3.0
BET	26.5	22.2	0.19	16.1	4.3
Bitcoin	12.0	66.6	0.71	43.1	19.2

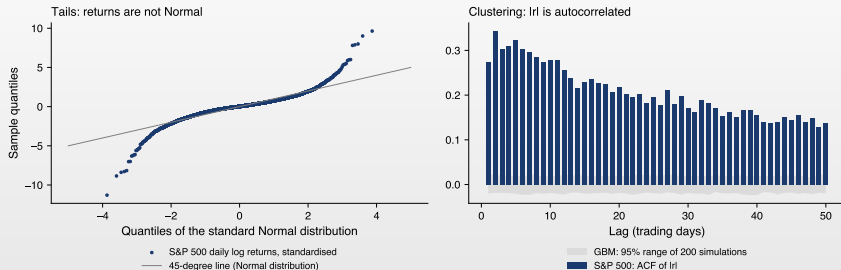
- Volatility is pinned down to 0.1–0.7 percentage points, the log drift only to 3–19 points; S&P 500: $\hat{\mu} = 8.3\% + 1.62\% = 10.0\%$ per year
- These s.e. are conditional on GBM: with excess kurtosis k , even independent returns inflate $\text{s.e.}(\hat{\sigma})$ by $\sqrt{(k+2)/2} = 2.5$ (S&P 500); block-bootstrap s.e. in Seminar 11, B2

How Precisely Can We Estimate the Drift?



- Left: half-width of the 95% CI for the drift against years of data; ± 1 point needs $(1.96 \hat{\sigma} / 0.01)^2 = 1243$ years for the S&P 500 and 17028 years for Bitcoin
- Right: S&P 500 on rolling 10-year windows: the log drift ranges from -6.0% (window ending 2009-03) to 15.8% (2000-09), each with a CI of about ± 11 points, so the variation is mostly noise; volatility stays between 14.1% and 22.3% : precise, but not constant
- Option pricing in [Chapter 12](#) avoids μ : under the risk-neutral measure only σ enters (Girsanov, Section 8)

Does GBM Fit the S&P 500?



- Left: quantile–quantile (QQ) plot of standardised daily log returns against the Normal distribution: excess kurtosis 10.9 instead of 0, worst day at -11.3 standard deviations, 62 days beyond ± 4 against 0.59 expected; Jarque–Bera (JB) statistic 45 805
- Right: autocorrelation function (ACF) of $|r_t|$: 0.27 at lag 1 and still 0.14 at lag 50, while 200 GBM samples of the same length never exceed 0.024 in absolute value
- GBM fails on both stylised facts of [Chapter 1 Cont \(2001\)](#); the rest of the chapter adds mean reversion, jumps and random volatility

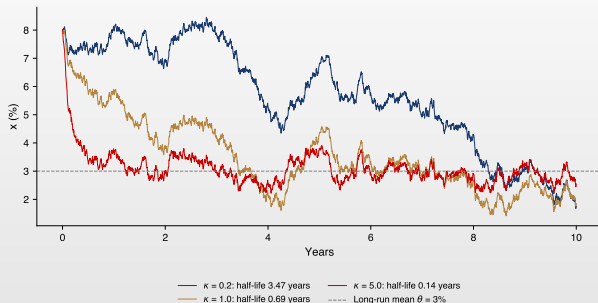
The Ornstein–Uhlenbeck (OU) Process

- SDE: $dx_t = \kappa(\theta - x_t) dt + \sigma dW_t$, $\kappa > 0$ [Uhlenbeck & Ornstein \(1930\)](#): long-run mean θ , speed of mean reversion κ , volatility σ
 - ▶ Above θ the drift is negative, below θ positive: a spring pulling x back; used for interest rates [Vasicek \(1977\)](#), volatility, spreads, commodity prices
- Itô's lemma on $e^{\kappa t} x_t$: $x_t = \theta + (x_0 - \theta)e^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-s)} dW_s$, a Normal variable with
 - ▶ $\mathbb{E}[x_t | x_0] = \theta + (x_0 - \theta)e^{-\kappa t}$, $\text{Var}(x_t | x_0) = \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa t})$
 - ▶ Stationary distribution ($t \rightarrow \infty$): $N(\theta, \sigma^2/(2\kappa))$
- **Half-life** $\ln 2/\kappa$: the time for an expected deviation from θ to halve
 - ▶ $\kappa = 0.5$ per year: half-life 1.39 years; unlike Brownian motion, the variance stops growing

Exact Discretisation: an AR(1)

- Sampled at step Δt the OU process is exactly an autoregressive model of order 1 (AR(1)):
 - ▶ $x_{t+\Delta t} = a + b x_t + \varepsilon_t$, $b = e^{-\kappa \Delta t}$, $a = \theta(1 - b)$, $\text{Var}(\varepsilon_t) = \frac{\sigma^2}{2\kappa}(1 - b^2)$
- Maximum likelihood = ordinary least squares (OLS) of $x_{t+\Delta t}$ on x_t , then invert the mapping
 - ▶ $\hat{\kappa} = -\ln \hat{b} / \Delta t$, $\hat{\theta} = \hat{a} / (1 - \hat{b})$, $\hat{\sigma} = \hat{s}_\varepsilon \sqrt{2\hat{\kappa} / (1 - \hat{b}^2)}$
- Standard errors for $\hat{\kappa}$ and $\hat{\theta}$ by the delta method from the OLS covariance of $(\hat{a}, \hat{b})$ (Seminar 11, A9)
- With daily data b is very close to 1: κ is hard to estimate, as with unit-root tests

Three Speeds of Mean Reversion

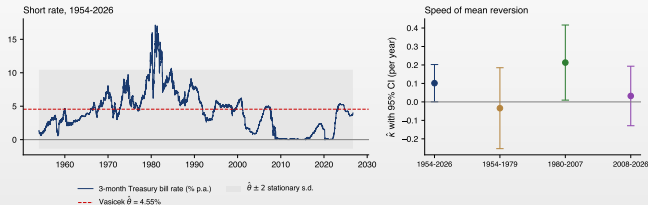


- OU paths driven by the same shocks, starting at 8%, with $\theta = 3\%$, $\sigma = 1\%$ and $\kappa = 0.2, 1, 5$; exact simulation, daily steps
- Half-lives 3.47, 0.69 and 0.14 years: with $\kappa = 5$ the gap from 8% to 3% closes within a few months
- Stationary standard deviations $\sigma / \sqrt{2\kappa}$: 1.58%, 0.71%, 0.32%; slow reversion looks like a random walk for years
- Long-horizon forecasts converge to θ

Short-Rate Models: Vasicek and CIR

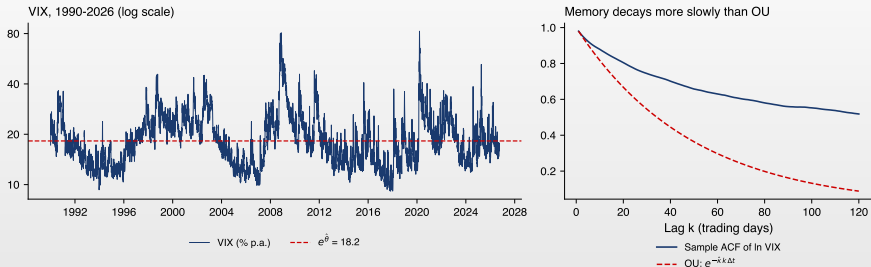
- **Vasicek**: the short rate r_t is an OU process [Vasicek \(1977\)](#)
 - ▶ Normal distribution: negative rates have positive probability $\Phi(-\theta\sqrt{2\kappa}/\sigma)$ in the long run
 - ▶ Bond prices and yields in closed form (Feynman–Kac, Section 8)
- **Cox–Ingersoll–Ross (CIR)**: $dr_t = \kappa(\theta - r_t)dt + \sigma\sqrt{r_t}dW_t$ [Cox, Ingersoll & Ross \(1985\)](#)
 - ▶ Volatility shrinks near zero; the rate stays positive if $2\kappa\theta \geq \sigma^2$ (Feller condition) [Feller \(1951\)](#)
- Which is closer to the data is an empirical question, tested in Section 7 [Chan, Karolyi, Longstaff & Sanders \(1992\)](#)

Vasicek on the 3-Month Treasury Bill Rate



- Left: daily 3-month US Treasury bill rate (DTB3), 1954–2026, with the Vasicek long-run mean $\hat{\theta} = 4.55\%$; $\hat{\kappa} = 0.101$ (s.e. 0.051), half-life 6.8 years, $\hat{\sigma} = 1.33\%$, daily $\hat{b} = 0.99960$: almost a unit root
- Right: $\hat{\kappa}$ with 95% CI by period: -0.034 (1954–1979, explosive), 0.213 (1980–2007), 0.032 (2008–2026); two of the three CIs include zero
- In finite samples $\hat{\kappa}$ overstates κ by roughly $4/T$ (T in years) [Tang & Chen \(2009\)](#); with the full-sample estimates as the truth, 500 samples of 72 years give a mean $\hat{\kappa} = 0.164$: bias 0.063 against $4/T = 0.055$, with 78% of the estimates above the truth
 - ▶ The bias depends on the calendar span, not on the sampling frequency: estimated half-lives are too short and mean reversion looks faster than it is (Seminar 11, B4)

Is Volatility Mean-Reverting? The VIX



- Left: VIX 1990–2026 on a log scale with the OU long-run level $e^{\hat{\theta}} = 18.2$; OU fit to $\ln VIX$: $\hat{\kappa} = 5.10$ (s.e. 0.53), half-life 34 trading days: spikes such as 82.7 (2020-03-16) fade within weeks
- Right: sample ACF of $\ln VIX$ against the OU-implied $e^{-\hat{\kappa}k\Delta t}$: at lag 60, 0.63 in the data against 0.30 under OU: a fast decay, then a slow component
- One OU factor is too simple: candidates are two-factor or long-memory models, while rough volatility (Section 12) is a distinct alternative; mean-reverting volatility is the building block of Heston

Infill versus Long-Span Asymptotics

Infill: $\Delta \rightarrow 0$ on a fixed $[0, T]$

- ▣ $\sum_k (\Delta X_k)^2 \rightarrow \int_0^T b_t^2 dt$
- ▣ The diffusion coefficient is identified without error in the limit: quadratic variation is observable (continuous Itô process, no measurement noise)
- ▣ Sampling more often helps σ (Section 5) and $b(x)$ (nonparametric, below)

Long span: $T \rightarrow \infty$

- ▣ The drift is not consistently estimable on a fixed span: its Fisher information grows with T , not with n
- ▣ GBM: $\text{s.e.}(\hat{m}) = \sigma/\sqrt{T}$; OU: $\text{Var}(\hat{\kappa}) \approx (e^{2\kappa\Delta} - 1)/(T\Delta) \rightarrow 2\kappa/T$ as $\Delta \rightarrow 0$ (Seminar 11, A9)
- ▣ The bias of $\hat{\kappa}$ does not shrink with finer sampling [Yu \(2012\)](#); $b = e^{-\kappa\Delta} \rightarrow 1$, so $\hat{\kappa}$ is far from Normal when κT is small

- ▣ One phenomenon, not two facts: the imprecise GBM drift and the biased $\hat{\kappa}$ are both long-span problems

Likelihood for Nonlinear Diffusions

- **Euler pseudo-likelihood:** a Gaussian transition $r_{t+\Delta} \mid r_t \sim N(r_t + a(r_t)\Delta, b^2(r_t)\Delta)$
 - ▶ Biased at order Δ : small with daily data, larger with monthly or quarterly steps
- **Exact CIR:** $2c r_{t+\Delta} \mid r_t$ is noncentral χ^2 with $4\kappa\theta/\sigma^2$ degrees of freedom [Cox, Ingersoll & Ross \(1985\)](#)
 - ▶ $c = 2\kappa/(\sigma^2(1 - e^{-\kappa\Delta}))$, noncentrality $2c r_t e^{-\kappa\Delta}$; Feller condition $\Leftrightarrow$ degrees of freedom ≥ 2 (Seminar 11, A5)
- General diffusions: closed-form Hermite expansions of the transition density [Aït-Sahalia \(2002\)](#)
- CKLS (Chan–Karolyi–Longstaff–Sanders): $dr = \kappa(\theta - r)dt + \sigma r^\gamma dW$ nests Vasicek ($\gamma = 0$) and CIR ($\gamma = 1/2$) [Chan, Karolyi, Longstaff & Sanders \(1992\)](#)
 - ▶ Gaussian (Euler) quasi-likelihood; tests on γ with sandwich standard errors whose score covariance is heteroskedasticity- and autocorrelation-consistent (HAC) [Newey & West \(1987\)](#)

Vasicek, CIR and CKLS on the T-Bill Rate, 1954–2007

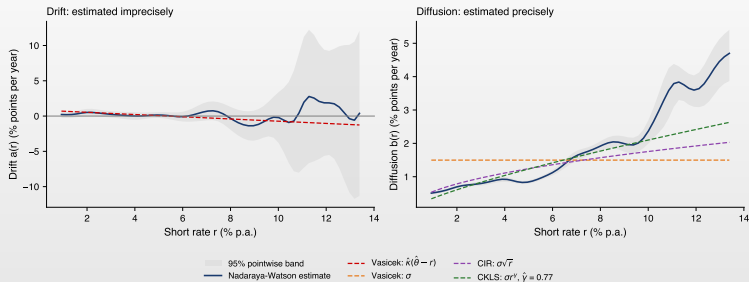
Model	$\hat{\kappa}$ daily (s.e.)	$\hat{\kappa}$ monthly (s.e.)	γ
Vasicek, exact	0.158 (0.073)	0.201 (0.083)	0
Vasicek, Euler	0.158 (0.073)	0.199 (0.082)	0
CIR, exact	0.113 (0.060)	0.153 (0.070)	1/2
CIR, Euler	0.093 (0.060)	0.144 (0.069)	1/2
CKLS, Euler	0.080 (0.057)	0.144 (0.068)	0.77 (0.033); 0.60 (0.149)

- Daily DTB3 ($n = 13\,487$) and month-end values ($n = 647$); 2008–2026 excluded: exact zeros and negative quotes lie outside the domain of these likelihoods (▶ [Appendix](#))
- Exact and Euler differ by less than one standard error: discretisation error is small next to the long-span uncertainty of $\hat{\kappa}$
- Volatility rises with the level: HAC Wald tests reject $\gamma = 0$ at both frequencies and $\gamma = 1/2$ with daily but not with monthly data (statistics: ▶ [Appendix](#))

Nonparametric Drift and Diffusion

- No functional form: kernel averages of the increments around each level x
 - ▶ $\hat{a}(x) = \sum_t w_t(x) \Delta r_t / \Delta$, $\hat{b}^2(x) = \sum_t w_t(x) (\Delta r_t)^2 / \Delta$, $w_t(x) = K((r_t - x)/h) / \sum_s K((r_s - x)/h)$
 - ▶ Nadaraya–Watson weights summing to one, Gaussian kernel K , bandwidth h [Stanton \(1997\)](#)
- $\mathbb{E}[\Delta r \mid r = x] / \Delta = a(x) + O(\Delta)$ and $\mathbb{E}[(\Delta r)^2 \mid r = x] / \Delta = b^2(x) + O(\Delta)$: exact only as $\Delta \rightarrow 0$
 - ▶ The diffusion converges at the infill rate, the drift only at the long-span rate; valid for non-stationary but recurrent processes [Bandi & Phillips \(2003\)](#)
- Caution: nonparametric specification tests reject far too often with persistent data [Pritsker \(1998\)](#)

Nonparametric Drift and Diffusion of the T-Bill Rate



- Daily DTB3, 1954–2007; bandwidth $h = 0.44$ percentage points; 95% heteroskedasticity-robust pointwise bands (serial dependence ignored)
- Drift: zero lies inside the band at 97% of the levels; diffusion: median relative s.e. 8%, $b(r)$ rises from 0.5% to 4.7% (log-slope 0.90)
- No parametric shape fits everywhere: inside the band at 35% (CKLS), 7% (CIR), 5% (Vasicek) of the levels, a descriptive share

Girsanov's Theorem

- Real-world measure P and an equivalent measure Q (the same events of probability zero), linked by a **Radon–Nikodym density**
 - ▶ $\frac{dQ}{dP} \Big|_{\mathcal{F}_T} = Z_T = \exp \left(- \int_0^T \theta_s dW_s - \frac{1}{2} \int_0^T \theta_s^2 ds \right)$
- **Girsanov (1960)**: under Q , $\tilde{W}_t = W_t + \int_0^t \theta_s ds$ is a Brownian motion if $\mathbb{E} \left[e^{\frac{1}{2} \int_0^T \theta_s^2 ds} \right] < \infty$ (Novikov condition)
- For $dX = a dt + b dW$: $dX = (a - b\theta) dt + b d\tilde{W}$
 - ▶ **Only the drift changes**: the diffusion coefficient and the quadratic variation are the same under P and Q
 - ▶ Under GBM (constant σ), realised variance ([Chapter 9](#)) measures the σ that prices options, while μ drops out; with stochastic volatility, P and Q weigh future variance paths differently, so historical variance is a benchmark, not a calibration

Market Price of Risk: P versus Q

- GBM: $\theta = (\mu - r)/\sigma$ is the **market price of risk**; under Q , $dS = rS dt + \sigma S d\tilde{W}$ and $e^{-rt}S_t$ is a Q -martingale (Seminar 11, A1)
 - ▶ S&P 500, 1990–2026: $\hat{\theta} = (10.0\% - 2.7\%)/18.0\% = 0.40$, s.e. $1/\sqrt{T} = 0.17$ (r : mean 3-month T-bill rate): as imprecise as the drift
- Vasicek with risk premium $\lambda\sigma$: $\kappa^Q = \kappa$, $\kappa\theta^Q = \kappa\theta - \lambda\sigma$
 - ▶ The long-run level that prices bonds differs from the one estimated on data (Section 6)
- Heston with volatility risk premium λv_t : $\kappa^Q = \kappa + \lambda$, $\kappa^Q\theta^Q = \kappa\theta$
 - ▶ $\lambda < 0$ (investors pay for protection against variance) gives $\theta^Q > \theta$: the variance risk premium of Chapter 9

Feynman–Kac: From Expectations to the Pricing PDE

□ If $dX = a(X) dt + b(X) d\tilde{W}$ under Q Kac (1949)

▶ $u(t, x) = \mathbb{E}^Q \left[e^{-\int_t^T r_s ds} g(X_T) \mid X_t = x \right]$ solves $u_t + a u_x + \frac{1}{2} b^2 u_{xx} - r u = 0$, $u(T, x) = g(x)$

□ Proof idea: $e^{-\int_0^t r_s ds} u(t, X_t)$ is a Q -martingale, so Itô's lemma must give it a zero drift

□ Vasicek bond ($g = 1$, $X = r$): $p(t, r; T) = \exp(A(\tau) - B(\tau)r)$, $\tau = T - t$ (Seminar 11, A3)

▶ $B(\tau) = \frac{1 - e^{-\kappa\tau}}{\kappa}$, $A(\tau) = \left(\theta^Q - \frac{\sigma^2}{2\kappa^2} \right) (B(\tau) - \tau) - \frac{\sigma^2 B(\tau)^2}{4\kappa}$

▶ Yields $y(\tau) = (B(\tau)r - A(\tau))/\tau$ tend to $\theta^Q - \sigma^2/(2\kappa^2)$: rate uncertainty lowers long yields (convexity);
comparison with Treasury yields: ▶ [Appendix](#)

□ Chapter 12 derives Black–Scholes by the same route

Merton's Jump-Diffusion

- Brownian paths are continuous, but prices gap on news: on 19 October 1987 the Dow Jones fell 22.6%, about 20 standard deviations under GBM with 18% volatility
 - ▶ A jump cannot be hedged by continuous trading [Merton \(1976\)](#)
- Log price = Brownian motion with drift + **compound Poisson** jumps: $d \ln S_t = m dt + \sigma dW_t + dJ_t$,
 $J_t = \sum_{i \leq N_t} Y_i$
 - ▶ N_t Poisson with intensity λ ; sizes $Y_i \sim N(\mu_J, \sigma_J^2)$ i.i.d.; W , N and Y independent; five parameters $(m, \sigma, \lambda, \mu_J, \sigma_J)$
- Return over Δt given k jumps: $N(m\Delta t + k\mu_J, \sigma^2\Delta t + k\sigma_J^2)$; unconditionally a **Poisson mixture** of Normal densities, i.i.d. over time
 - ▶ $f(r) = \sum_{k \geq 0} e^{-\lambda\Delta t} \frac{(\lambda\Delta t)^k}{k!} \varphi(r; m\Delta t + k\mu_J, \sigma^2\Delta t + k\sigma_J^2)$
- Exact simulation: draw $N \sim \text{Poisson}(\lambda\Delta t)$, then draw from the Normal distribution with the conditional moments

Moments of the Merton Return

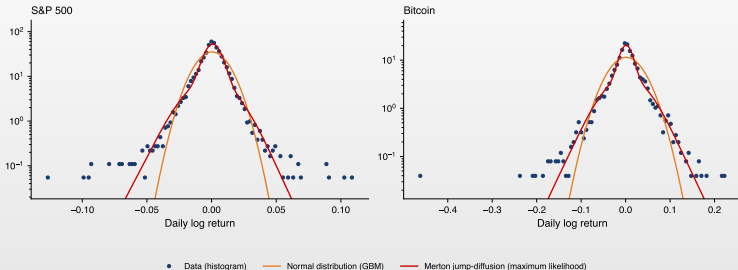
- Cumulants of the return over Δt (the k -th cumulant of a compound Poisson sum is $\lambda \Delta t \mathbb{E}[Y^k]$):
 - ▶ $\kappa_2 = \sigma^2 \Delta t + \lambda \Delta t (\mu_J^2 + \sigma_J^2)$, $\kappa_3 = \lambda \Delta t (\mu_J^3 + 3\mu_J \sigma_J^2)$
 - ▶ $\kappa_4 = \lambda \Delta t (\mu_J^4 + 6\mu_J^2 \sigma_J^2 + 3\sigma_J^4)$; skewness $\kappa_3 / \kappa_2^{3/2}$, excess kurtosis κ_4 / κ_2^2
- Negative μ_J gives negative skewness; any jump gives positive excess kurtosis
- Excess kurtosis falls like $1/\Delta t$ as the horizon grows: jumps matter most at short horizons
 - ▶ Example: $\sigma = 15\%$, $\lambda = 5$, $\mu_J = -3\%$, $\sigma_J = 4\%$: daily excess kurtosis 17.6, skewness -2.07 (Seminar 11, A7)

Maximum Likelihood Estimates

	$\hat{\sigma}$ (%)	$\hat{\lambda}$ (per year)	$\hat{\mu}_J$ (%)	$\hat{\sigma}_J$ (%)	Jump share of variance	LR
S&P 500	9.4	102 (11)	-0.16	1.44	71%	2 401
BET	11.3	79 (7)	-0.09	2.07	73%	2 326
Bitcoin	24.6	239 (21)	0.00	3.85	85%	1 357

- Log-likelihood maximised numerically over the Poisson mixture (up to 12 jumps per day); standard errors from the Hessian in brackets
- The likelihood is **unbounded**: with $m\Delta t$ equal to one observed return and $\sigma \rightarrow 0$ that density explodes, so the MLE is a local maximum [Honoré \(1998\)](#)
 - The LR test against GBM ($\lambda = 0$) is defined on $\sigma \geq 5\%$ and is not χ^2_3 : $\lambda = 0$ lies on the boundary and μ_J, σ_J are unidentified; bootstrap critical values in Seminar 11, B6
- The fit prefers **many small** jumps (about 80–240 per year), not a few crashes

Merton Fit to the Return Distribution

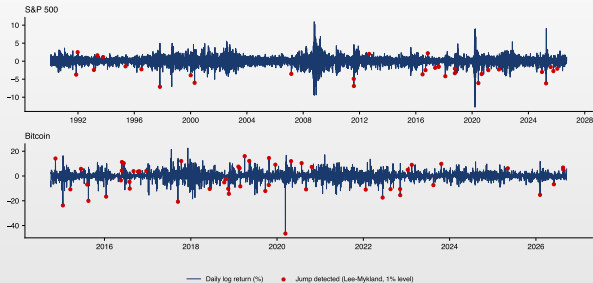


- Density of daily log returns on a log scale: histogram, the Normal distribution implied by GBM and the fitted Merton mixture; Merton follows the peak and the shoulders far better, and the LR statistics in the thousands reject GBM
- S&P 500: implied excess kurtosis 3.7 against 10.9 in the data, skewness -0.31 against -0.36 ; the most extreme days lie beyond both fitted curves
- About 102 “jumps” a year: the model uses jumps to mimic **changing volatility**; an i.i.d. model cannot tell a jump from a volatile period, so a direct test is needed

Detecting Jumps: the Lee–Mykland Test

- Idea: a jump is a return that is huge **relative to the local volatility** Lee & Mykland (2008)
 - ▶ $L_t = r_t / \hat{\sigma}_t$, $\hat{\sigma}_t^2 = \frac{1}{K-2} \sum_{j=t-K+2}^{t-1} |r_j| |r_{j-1}|$ (bipower variation, robust to jumps)
- Under no jumps, the maximum of $|L_t|$ over n days has a Gumbel limit
 - ▶ Reject if $(|L_t| - C_n) / S_n > -\ln(-\ln(1 - \alpha))$; at $\alpha = 1\%$ this threshold for the **normalised** statistic, not for $|L_t|$, is 4.60
 - ▶ $C_n = \frac{\sqrt{2 \ln n}}{c} - \frac{\ln \pi + \ln \ln n}{2c\sqrt{2 \ln n}}$, $S_n = \frac{1}{c\sqrt{2 \ln n}}$, $c = \sqrt{2/\pi}$ (the mean of $|Z|$, which rescales the bipower estimate), n = number of tested days
- Daily data: window $K = 16$ days, as recommended by the authors
- Controls, asymptotically, the probability of a false jump over the **whole** sample, not day by day

Jumps Detected in the S&P 500 and Bitcoin



	S&P 500	Bitcoin
Candidates	32	52
Per year	0.87	4.35
Negative (%)	84	52
In the top 10 moves	0	5

- Daily log returns with the days flagged at the 1% level
- S&P 500: the largest moves came in turbulent periods (2008, 2020) and are not flagged, while a 1.2% move in a calm market is (1993-10-14)
- Candidate detections are not the fitted Merton intensity: the gap may reflect stochastic volatility, drift bursts (short explosive trends in the drift) and limited detection power in daily data

 MFM_ch11_jumps

The Heston Model

- Two SDEs, for the log price and for the instantaneous variance v_t [Heston \(1993\)](#)
 - ▶ $d \ln S_t = (\mu - \frac{1}{2} v_t) dt + \sqrt{v_t} dW_t^{(1)}$
 - ▶ $dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^{(2)}, \text{Corr}(dW^{(1)}, dW^{(2)}) = \rho$
- Variance follows a CIR process: mean-reverting to θ with speed κ ; ξ = volatility of volatility
 - ▶ Feller condition $2\kappa\theta \geq \xi^2$: variance never touches zero [Feller \(1951\)](#)
- $\rho < 0$: **leverage effect**, falling prices come with rising volatility
 - ▶ Simulation: Euler scheme with full truncation $v^+ = \max(v, 0)$ [Lord, Koekkoek & van Dijk \(2010\)](#); an alternative discretisation, the quadratic-exponential scheme [Andersen \(2008\)](#)

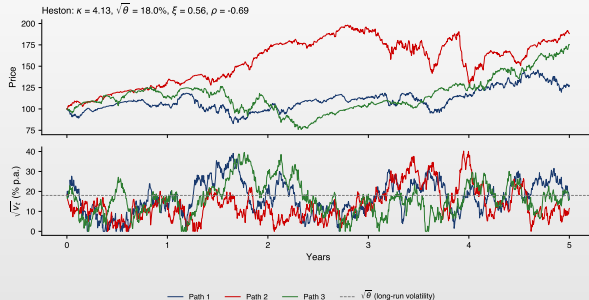
Heston Parameters from the VIX

- Proxy $y_t = (\text{VIX}_t/100)^2$ in place of v_t ; discretised CIR regression without an extra intercept
 - ▶ $\frac{y_{t+\Delta t} - y_t}{\sqrt{y_t}} = \kappa\theta \frac{\Delta t}{\sqrt{y_t}} - \kappa\Delta t\sqrt{y_t} + \xi\sqrt{\Delta t}\varepsilon_t$; ρ = correlation of return and variance shocks
- S&P 500 and VIX, 1990–2026: $\hat{\kappa} = 4.13$ (half-life 42 days), $\hat{\xi} = 0.56$, $\hat{\rho} = -0.69$
 - ▶ $\sqrt{\hat{\theta}} = 20.9\%$ from the VIX, but the realised volatility is 18.0%; Feller ratio of the proxy $2\hat{\kappa}\hat{\theta}/\hat{\xi}^2 = 1.16$
- But y_t is **not** v_t : it is a risk-neutral expectation of future variance, affine in v_t (next slide)
 - ▶ For real-world paths we set θ to the realised variance; $\hat{\xi}$ is attenuated relative to ξ

What the VIX Identifies in Heston

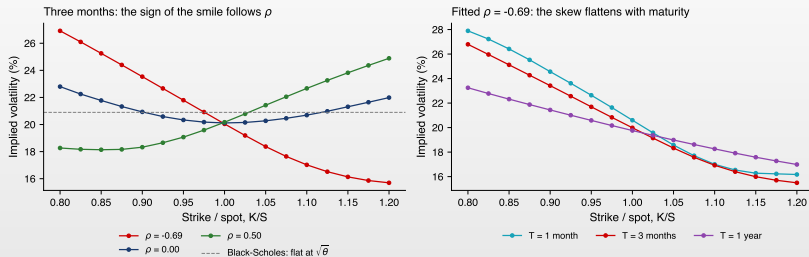
- Under Q : $y_t = \frac{1}{\tau} \mathbb{E}^Q \left[\int_t^{t+\tau} v_s ds \mid \mathcal{F}_t \right] = a + b v_t$, $\tau = 30/365$ (Seminar 11, A6)
 - ▶ $b = \frac{1 - e^{-\kappa^Q \tau}}{\kappa^Q \tau} < 1$, $a = \theta^Q(1 - b)$: **affine** in v_t , with Q -parameters
- What the regression on y_t recovers
 - ▶ κ : the P -speed, since an affine map keeps the dynamics of v_t ; ρ : unchanged, since correlation is scale-free, but estimating it needs paired return and variance shocks, and daily proxy changes only approximate them
 - ▶ ξ : attenuated; the diffusion of y is $\xi \sqrt{b(y - a)}$, so $\xi_y = b \xi \sqrt{v/(a + bv)}$ varies with the state and equals $b \xi$ only at $v_t = \theta^Q$ (local correction $\hat{\xi}/b$: ▶ Appendix)
 - ▶ The long-run level $a + b \theta^P$ mixes θ^Q and θ^P : recover $\theta^P = (\theta_y - a)/b$ before a Feller check on v_t
- Proper estimation fits the shifted square-root dynamics of y , or v_t and the observation equation jointly
- κ^Q is not identified from the VIX alone: joint P/Q estimation with option prices [Chernov & Ghysels \(2000\)](#), or moments of realised variance [Bollerslev & Zhou \(2002\)](#)

Simulated Heston Paths



- Three five-year paths with the VIX-based $\hat{\kappa}$, $\hat{\xi}$ (uncorrected), $\hat{\rho}$ and θ equal to the realised S&P 500 variance; top: price, bottom: $\sqrt{v_t}$
- Clustering now comes from the model; price falls coincide with volatility spikes (simulated return–variance correlation -0.69 , the leverage effect through ρ)
- Feller ratio $2\hat{\kappa}\theta/\hat{\xi}^2 = 0.86 < 1$: the CIR variance can reach zero; negative values arise only in the Euler approximation, and full truncation sets them to zero (1.4% of the steps)

A Preview of Chapter 12: the Volatility Smile



- Implied volatility: the σ that makes the Black–Scholes price equal to a given option price, flat across strikes under GBM; here from Heston call prices (Monte Carlo, 100,000 antithetic paths)
- Left, three months: with $\hat{\rho} = -0.69$, 26.9% at strike 80% of spot, 20.0% at the money, 15.7% at 120%: a downward **skew**; with $\rho = 0$ a symmetric **smile** (22.8%, 20.1%, 22.0%)
- Right: the skew flattens with maturity, from 27.9% – 16.2% at one month to 23.2% – 17.0% at one year; illustration only: pricing needs κ^Q , θ^Q calibrated to option prices ([Chapter 12](#))

Affine Models and the GARCH Diffusion Limit

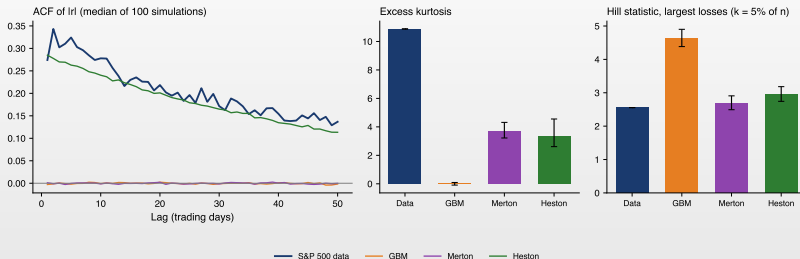
Affine jump-diffusions Duffie, Pan & Singleton (2000)

- ▣ Drift, variance and jump intensity affine in the state X_t : Vasicek, CIR, Merton, Heston and Bates
- ▣ $\mathbb{E} \left[e^{iu^\top X_\tau} \mid X_t \right] = \exp \left(\alpha(\tau) + \beta(\tau)^\top X_t \right)$, with α, β solving Riccati ordinary differential equations
- ▣ Option prices by Fourier inversion; the Vasicek $A(\tau)$, $B(\tau)$ of Section 8 are the simplest case

From GARCH to continuous time Nelson (1990)

- ▣ As the step shrinks, GARCH(1,1) converges to $dv_t = \kappa(\theta - v_t) dt + \xi v_t dW_t^{(2)}$: like Heston, with v_t instead of $\sqrt{v_t}$
- ▣ $\kappa \approx (1 - \alpha - \beta)/\Delta t$: $\alpha + \beta = 0.99$ with daily data gives $\kappa \approx 2.5$ per year, a half-life of about 70 days ([Chapter 5](#))
- ▣ GARCH and its limit are not asymptotically equivalent experiments: inference can differ even as $\Delta t \rightarrow 0$ [Wang \(2002\)](#)

Which Model Reproduces the S&P 500?



- 100 simulated samples of 9 245 days from each fitted model: median ACF of $|r_t|$, excess kurtosis and the Hill statistic of the largest losses ($k = 5\%$ of the returns; lower = heavier tail), with 90% ranges; data: kurtosis 10.88, Hill 2.56
- Merton gets the Hill statistic (2.70) and skewness (-0.32 against -0.36) but no clustering; Heston, with the VIX-based parameters, gets the clustering but no skewness (0.00)
- No fitted model reaches the kurtosis of the data; the evidence of Andersen, Benzoni & Lund (2002) and Eraker, Johannes & Polson (2003) favours jumps and stochastic volatility (SVJ, Bates (1996)); full table in the [Appendix](#)

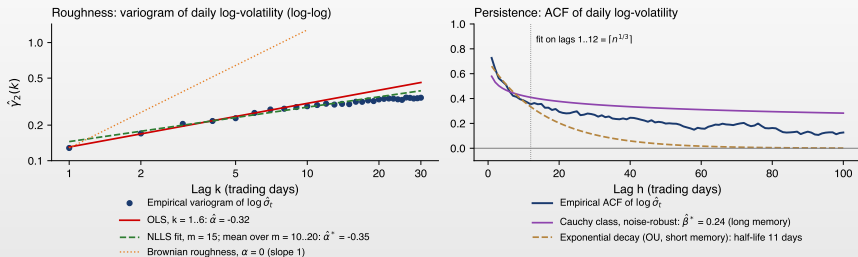
Which Model When? Features, Identification and Limits

Feature in the data	Model	What the data identify	Unresolved
Positive prices, $\sqrt{t}$ scaling	GBM	σ precisely; μ only over very long spans	fat tails, clustering
Mean reversion of rates and of the VIX	OU, Vasicek, CIR, CKLS	$b(r)$ precisely; κ biased upwards, needs a long span	one factor too simple for the VIX
Gaps, short-horizon fat tails	Merton	λ, σ_J only under i.i.d. returns; the jump share depends on the frequency	no clustering
Clustering, leverage, implied-volatility skew	Heston	κ, ρ from the VIX; ξ attenuated; $\theta^Q \neq \theta^P$	kurtosis and skewness of the data
All of the above	Bates (Heston + jumps)	returns and option surfaces jointly	latent states: filtering needed

- Roughness and long memory of volatility, next, are features none of these one-factor models produces

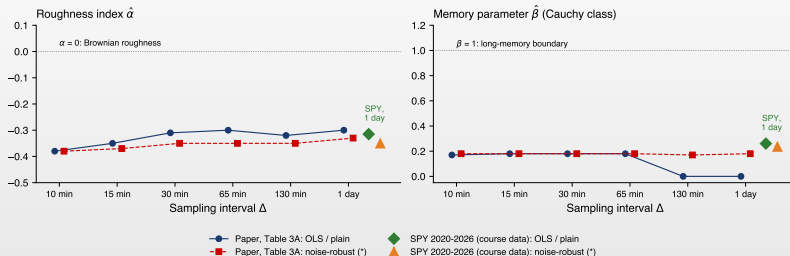
Research Question: Rough, Yet Persistent?

- Paper: [Bennedsen, Lunde & Pakkanen \(2022\)](#), *Journal of Financial Econometrics*
 - ▶ It builds on [Gatheral, Jaisson & Rosenbaum \(2018\)](#): log-volatility as a fractional OU process with Hurst exponent $H < 1/2$ (RFSV), whereas the long-memory model of [Comte & Renault \(1998\)](#) uses $H > 1/2$ (Introduction)
- Roughness index α : $1 - \rho(h) \sim c|h|^{2\alpha+1}$ as $h \rightarrow 0$, with ρ the autocorrelation of log-volatility (eq. (1.1)); $\alpha < 0$: paths rougher than Brownian motion
 - ▶ Memory parameter β : $\rho(h) \sim c|h|^{-\beta}$ as $h \rightarrow \infty$ (eq. (2.2)); $\beta < 1$: the ACF is not integrable, i.e. long memory
- The problem: in the fractional OU process a single H fixes both, $\alpha = H - 1/2$ and $\beta = 2 - 2H$ (Section 1): with $H < 1/2$, $\beta > 1$, so a rough fractional OU process cannot have long memory
- Research questions (Introduction, Sections 4 and 5)
 - ▶ are realised volatilities both rough and persistent?
 - ▶ do models that set α and β separately forecast better?
- Here: the daily part of the analysis on SPY; the model, the estimators and the forecast results of the paper are in the [Appendix](#), the intraday extension in Seminar 11, C3

Case Study: Replication on SPY, $\Delta = 1$ Day

- SPY 5-minute bars, 1 468 full days (October 2020 – September 2026); $x_t = \log \hat{\sigma}_t = \frac{1}{2} \log BV_t$, daily bipower variation of the 78 five-minute returns [Barndorff-Nielsen & Shephard \(2004\)](#)
- Left, rough: variogram $\hat{\gamma}_2(k) = \frac{1}{n-k} \sum_t (x_{t+k} - x_t)^2$; $\hat{\alpha}_{OLS} = -0.32$ ($k = 1, \dots, 6$), noise-robust $\hat{\alpha}_{NLLS}^* = -0.35$; the paper at 1 day: $-0.30, -0.33$ (Table 3, Panel A)
- Right, persistent: ACF 0.13 at lag 100; Cauchy $\hat{\beta} = 0.26$, $\hat{\beta}^* = 0.24$ on 12 lags, below 1: consistent with long memory, not a proof; an OU fit (half-life 11 days) implies 0.001 at lag 100

Case Study: SPY Against Table 3 of the Paper



- Lines: E-mini S&P 500 futures, 2011–2014, Δ from 10 minutes to 1 day (Bennedsen, Lunde & Pakkanen (2022), Table 3, Panel A); markers: SPY at 1 day, 2020–2026, course data; * = noise-robust
- $\hat{\alpha}$ barely moves with Δ (−0.38 to −0.30): roughness is a property of the process, and SPY falls in the same band; $\hat{\beta}^* = 0.17$ –0.18 at every Δ , far below 1, while the plain $\hat{\beta}$ is unstable
- Limits: another contract, period and estimator (bars, no pre-averaging); realised measures estimate integrated, not spot, variance

An Open Question: How Much of the Price Path Is Discontinuous?

- Model: $dX_t = \mu_t dt + \sigma_t dW_t + J_t dN_t$; the answer depends on the sampling frequency and on the model
 - ▶ Daily Merton MLE (this chapter): jumps carry 71% of S&P 500 variance; the 5-minute bipower test (Chapter 9): 2.8% for SPY
 - ▶ Christensen, Oomen & Podolskij (2014): at tick frequency most jumps detected at coarser sampling disappear; they are rapid but continuous moves
 - ▶ Christensen, Oomen & Renò (2022): **drift bursts**, short explosive trends in μ_t , look like jumps in coarse data
- Formal question: $H_0: \lambda = 0$ (continuous path with stochastic volatility) against $H_1: \lambda > 0$
 - ▶ Non-standard: $\lambda = 0$ lies on the boundary and (μ_J, σ_J) are unidentified under H_0 , so the LR statistic is not χ^2

The Discovery Loop and What the Researcher Checks

The loop with an AI assistant

- Assistant: an LLM such as Claude, ChatGPT or Gemini; **1. Literature:** tabulate published jump shares by frequency; check every DOI on [Crossref](#)
- **2. Hypotheses:** Merton jumps, Heston volatility, drift bursts, each with a distinct testable implication
- **3. Replicate first:** $\hat{\lambda} = 102$ per year, jump share 71%
- **4. Pre-registered robustness, 5. AI as a hostile referee, 6. Report** every specification, the code and the AI log (prompts: [Appendix](#))

Checks: failure and test

- Boundary, unidentified parameters: parametric bootstrap under H_0
- Look-ahead in the volatility filter: standardise only with $\mathcal{F}_{t-1}$
- Frequency mismatch: jump shares at every frequency on a common sample
- Multimodal likelihood: many starting points and a profile in λ
- Invented numbers: every claim needs a table and a DOI

Mini-Case: Jumps or Stochastic Volatility in Disguise?

- Replication: Merton on S&P 500 daily returns, $\hat{\lambda} = 102$ per year, jump share 71% (Section 9)
- A typical assistant proposal: “71% of variance comes from jumps”. Rival hypothesis: the Poisson mixture absorbs time-varying volatility; then standardising by EWMA volatility ($\lambda_{\text{EWMA}} = 0.94$, known at $t - 1$) removes the jumps

	Raw returns	EWMA-standardised
Excess kurtosis	10.9	2.9
LR, Merton vs the Normal distribution	2 401	619
$\hat{\lambda}$ per year (s.e.)	102 (11)	124 (53)
Jump share of variance	71%	51%

- Standardising removes most kurtosis, yet the model still reports a “jump” about every other day: $\hat{\lambda}$ measures fat tails, not discontinuities
- Interpretation: daily data do not identify the jump share; Lee–Mykland finds 0.87 jumps per year and 5-minute data give 2.8%, a different frequency and sample. Code: the chapter’s Quantlet (AI case study)

Your Turn: A Project Seed

- Questions
 - ▶ how much of S&P 500 and Bitcoin variance is discontinuous?
 - ▶ how does this share depend on the sampling frequency?
 - ▶ are the detected “jumps” drift bursts?
- Replicate first: Merton jump share 71% (daily S&P 500), Lee–Mykland 0.87 jumps per year, bipower jump share 2.8% (SPY, 5 minutes, [Chapter 9](#))
- Deliverables
 - ▶ A pre-registration (hypotheses, frequencies, filters, tests, bootstrap design) dated before the first run
 - ▶ The jump share at 5, 15, 30 minutes and daily; the drift-burst statistic of [Christensen, Oomen & Renò \(2022\)](#) around April 2025; a bootstrap LR test of Heston (null, no jumps) against Bates
 - ▶ Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Declare your AI use in `AI_USE.md` and the errors you caught in `AI_ERRORS.md`:
github.com/danpele/MFM/tree/main/project

Key Takeaways and Next: Chapter 12

- Brownian motion is the universal limit of random walks; its quadratic variation t drives Itô calculus
- Itô's lemma gives exact solutions (GBM, Ornstein–Uhlenbeck, Merton); simulate exactly, or with Euler–Maruyama and Milstein at a known error order
- Identification: infill pins down volatility; drift and mean-reversion speed need long spans, and $\hat{\kappa}$ is biased upwards
- Girsanov changes only the drift; Feynman–Kac turns a Q -expectation into a PDE; $(\text{VIX}/100)^2$ is a Q -expectation, affine in v_t
- In the fitted models jumps help with the tails and stochastic volatility with the clustering; volatility is rough and persistent, which one-factor models cannot produce
- Next, [Chapter 12](#): pricing options by replication and reading the volatility surface; code and data: github.com/danpele/MFM/tree/main/Quantlets/Ch_11

Appendix: From Brown to Bachelier, Einstein and Wiener



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons



Albert Einstein in 1905

Photo: Lucien Chavan / ETH Zürich (1905); public domain; Wikimedia Commons

- 1827: the botanist Robert Brown sees pollen particles move in an irregular zig-zag under the microscope
- Bachelier (1900): French government bond (rente) prices and their options follow a random walk in continuous time
 - ▶ The spread of the price grows with $\sqrt{t}$, five years before physics
- Einstein (1905): molecular collisions explain Brown's motion; the mean squared displacement is linear in t
- Wiener (1923): Brownian motion as a probability measure on continuous paths, the **Wiener process**

Appendix: Itô, Samuelson, Black, Scholes and Merton



Kiyosi Itô (1915–2008)

Photo: Konrad Jacobs (1970); CC
BY-SA 2.0 de; Wikimedia Commons



Paul A. Samuelson (1915–2009)

Photo: Bernard Gotfryd
(1970–1975); public domain;
Wikimedia Commons

- Itô (1944): the stochastic integral and its chain rule, the main working tool of this chapter
- Osborne (1959): log prices, not prices, behave like Brownian motion; Samuelson (1965): geometric Brownian motion
- Black & Scholes (1973) and Merton (1973): an option can be replicated in continuous time
- Merton (1976) adds jumps, Heston (1993) random volatility; textbook treatment: Franke, Härdle & Hafner (2019), Part I

Appendix: A Crash in Pictures, Amsterdam 1987



Amsterdam Stock Exchange, 21 October 1987, after the crash of 19 October

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia Commons

- 19 October 1987: the Dow Jones fell 22.6% in one day, about 20 standard deviations under GBM with 18% volatility
- Under a Normal distribution such a day would not be expected once in the age of the universe: prices gap, and continuous models need jumps or fat-tailed innovations
- A gap cannot be hedged by continuous trading, the argument of [Merton \(1976\)](#)

Appendix: Data Cleaning and Technical Conditions

- Data: equity indices on weekdays, stale quotes removed (unchanged close and no intraday range), genuine zero returns kept; Bitcoin 7 days a week
- Itô integral: left-point sums converge for regular integrands (e.g. continuous and adapted); a general H is reached by approximation with simple processes
 - ▶ If only $\int_0^T H_t^2 dt < \infty$ almost surely, $\int H dW$ is merely a local martingale and its mean need not be zero
- SDE: existence and uniqueness of a strong solution if a and b are Lipschitz in x with at most linear growth [Kloeden & Platen \(1992\)](#)
 - ▶ Square-root coefficients (CIR, Heston) are not Lipschitz at zero, yet have unique strong solutions (Yamada–Watanabe conditions)
- Donsker: convergence in distribution in the space of càdlàg paths with the Skorokhod topology [Billingsley \(1999\)](#)

Appendix: Short Rates, More Numbers

- DTB3, 1954–2026: the rate ranged from -0.05% (2020-03-26) to 17.1% (1980-12-11); 4.3% of the days were below 0.05%
 - ▶ The fitted Vasicek model gives a 6.1% long-run probability of a negative rate
 - ▶ 2008–2026: 13 exact zeros and 8 negative quotes, outside the domain of the CIR and CKLS likelihoods; 17% of the days form a near-zero regime below 0.05%
- CKLS tests on γ , 1954–2007, HAC Wald with 11 (daily) and 6 (monthly) lags
 - ▶ $\gamma = 0$ rejected: daily 545, monthly 16; $\gamma = 1/2$ rejected daily (67.0; 22.1 with 55 lags), not monthly (0.5 against 3.84)
 - ▶ The Gaussian LR for $\gamma = 1/2$ rejects monthly (10.1): it ignores volatility clustering; [Chan, Karolyi, Longstaff & Sanders \(1992\)](#) found $\hat{\gamma} \approx 1.5$ for 1964–1989
- Vasicek bond with the Section 6 estimates, $\lambda = 0$ and $r = 3.99\%$: 10-year yield 4.05% ; the 10-year Treasury yield (FRED DGS10) on 2026-09-18: 5.01%
 - ▶ The gap is a term premium ($\lambda \neq 0$) plus misspecification
- Nonparametric estimation: [Stanton \(1997\)](#) also estimates the market price of interest-rate risk; the specification test of [Aït-Sahalia \(1996\)](#) over-rejects with persistent data [Pritsker \(1998\)](#)



Appendix: The VIX Correction by κ^Q and the Full Model Comparison

κ^Q	1	2.5	5	10
b	0.96	0.90	0.82	0.68
Local correction $\hat{\xi}/b$	0.58	0.62	0.68	0.82

	Data	GBM	Merton	Heston
Excess kurtosis	10.88	0.00	3.66	3.30
Skewness	-0.36	0.00	-0.32	0.00
Hill statistic	2.56	4.64	2.70	2.95
ACF of $ r $, lag 1	0.27	0.00	0.00	0.28
ACF of $ r $, sum of lags 1–20	5.32	0.00	0.00	4.78

- $\hat{\xi}/b$ is exact only at $v_t = \theta^Q$. Model columns: medians of 100 simulations; Hill statistic of the data 2.97, 2.56, 2.04 for $k = 2.5\%$, 5% , 10% of the returns, a tail index only under regular (Pareto-type) variation
- A Heston model with a lower Feller ratio could reach the kurtosis: short-horizon excess kurtosis $\approx 3\xi^2/(2\kappa\theta)$

Appendix: Further Reading

- Heavier tails: Lévy processes and α -stable distributions [Pele \(2014\)](#)
 - ▶ Crypto-assets form a distinct class by the shape of their return distributions [Pele et al. \(2023\)](#)
 - ▶ High-frequency information about Bitcoin volatility and tail risk [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)
- Stochastic calculus: [Shreve \(2004\)](#), Chapters 5–6; [Franke, Härdle & Hafner \(2019\)](#), chapters on the Wiener process and stochastic integration
- Exercises with solutions on all models of this chapter: [Borak, Härdle & López-Cabrera \(2013\)](#); the Heston model: [Čížek, Härdle & Weron \(2011\)](#); applications: [Härdle, Chen & Overbeck \(2017\)](#)
- Monte Carlo methods: [Glasserman \(2003\)](#); numerical SDEs: [Kloeden & Platen \(1992\)](#)

Appendix: Case Study, Model, Data and Method

- **Model:** $\sigma_t = \xi e^{X_t}$ (eq. (1.3)), $X_t = \int_{-\infty}^t g(t-s) v_s dW_s$, a Brownian semistationary (BSS) process with kernel g and volatility-of-volatility process v_s (eq. (1.5)); the forecasts take $v_s \equiv 1$, a Gaussian X (Section 5)
 - ▶ Gamma kernel $g(x) = x^\alpha e^{-\lambda x}$ (Example 1.2): roughness α , short memory with exponential decay at rate λ
 - ▶ Power kernel $g(x) = x^\alpha (1+x)^{-\gamma-\alpha}$ (eq. (1.8)): $\beta = \gamma$ if $\gamma > 1$, $\beta = 2\gamma - 1$ if $\gamma \in (1/2, 1)$ (Remark 1.2); Gaussian alternative: the Cauchy class, $\rho(h) = (1 + |h|^{2\alpha+1})^{-\beta/(2\alpha+1)}$ (Section 1.1.1)
- **Step 1,** α from the variogram $\hat{\gamma}_2(k)$ of $\log \hat{\sigma}$: OLS of $\log \hat{\gamma}_2(k)$ on $\log k$, $k = 1, \dots, 6$, $\hat{\alpha} = (\hat{a}_1 - 1)/2$ (eq. (2.1))
 - ▶ Noise-robust NLLS: $\hat{\gamma}_2(k) = b_0 + b_1(k\Delta)^{2\alpha+1}$, the constant absorbs the measurement noise; average the estimates obtained with $m = 10, \dots, 20$ lags (Sections 2.1 and 3.1)
 - ▶ **Step 2:** β , γ or λ by matching the model ACF to the empirical one on the first $\lceil n^{1/3} \rceil$ lags (Sections 2.2 and 4.1)
- **Data:** E-mini S&P 500 futures, every trade, 2011–2014 (996 days), 09:30–16:00 New York time; $\hat{\sigma}_t^2 =$ pre-averaged bipower variation over Δ , divided by Δ , for Δ from 10 minutes to 1 day (Sections 4 and 4.1)
 - ▶ Intraday seasonality removed with the flexible Fourier form of [Andersen & Bollerslev \(1997\)](#); daily data on 1944 liquid US stocks, 2003–2013 (Sections 4.1 and 4.2)
- **Forecasts:** variance summed over the next $h = 1, 2, 5, 10, 20$ steps, rolling window of 200 observations, out of sample 2 January 2013 – 31 December 2014 (Section 5.1.1)
 - ▶ Rivals: random walk, AR(5), AR(10), ARMA(1,1), log-HAR (heterogeneous autoregressive model for log volatility) [Corsi \(2009\)](#), ARFIMA(0,d,0), RFSV; losses MSE and QLIKE; 90% and 75% MCS [Hansen, Lunde & Nason \(2011\)](#)

Appendix: Case Study, Estimators and Forecast Losses

- **Variogram** of $x_t = \log \hat{\sigma}_t$: $\hat{\gamma}_2(k) = \frac{1}{n-k} \sum_{t=1}^{n-k} (x_{t+k} - x_t)^2$; for small k , $\gamma_2(k) \approx c k^{2\alpha+1}$ (Section 2.1)
 - ▶ **Bipower variation** of the M returns r_j in a bin: $BV = \frac{\pi}{2} \sum_{j=2}^M |r_j| |r_{j-1}|$, robust to jumps [Barndorff-Nielsen & Shephard \(2004\)](#); the pre-averaged version first averages returns over short windows to damp microstructure noise (Appendix A of the paper)
- **Forecast object**: $FO_t(\Delta, h) = \sum_{k=1}^h BV_{t+k\Delta}$ (eq. (5.3)); for a Gaussian log model, $\mathbb{E}[e^{X_{t+\Delta}} | \mathcal{F}_t] = \exp(m + s^2/2)$, with m, s^2 the conditional mean and variance (eq. (5.2))
 - ▶ Losses over the out-of-sample origins: $MSE = \text{mean}(\widehat{FO}_t - FO_t)^2$, $QLIKE = \text{mean}(\log \widehat{FO}_t + FO_t/\widehat{FO}_t)$ (Section 5.1.1)
- **Model confidence set (MCS)** [Hansen, Lunde & Nason \(2011\)](#): a set of models that contains the best model with probability 90% (or 75%)
 - ▶ Membership is not a ranking: the models inside the MCS cannot be told apart at that level

Appendix: Case Study, Results in the Paper

log σ , E-mini S&P 500 (Table 3, Panel A)	$\hat{\alpha}_{OLS}$	$\hat{\alpha}_{NLLS}^*$	$\hat{\beta}_{Cauchy}^*$	$\hat{\lambda}_{Gamma}^*$
$\Delta = 10$ minutes	−0.38	−0.38	0.18	0.02
$\Delta = 65$ minutes	−0.30	−0.35	0.18	0.02
$\Delta = 1$ day	−0.30	−0.33	0.18	0.02

- Rough at every scale: $\hat{\alpha}$ between −0.38 and −0.30 from 10 minutes to 1 day (Table 3, Panel A; * = noise-robust): log-volatility is far rougher than Brownian motion ($\alpha = 0$)
 - ▶ And persistent: $\hat{\beta}_{Cauchy}^*$ between 0.17 and 0.18 at every Δ , below 1, and $\hat{\lambda}^*$ between 0.02 and 0.03 (Table 3, Panel A): long memory together with roughness
 - ▶ 1 944 US stocks: $\hat{\alpha}^*$ mostly between −0.45 and −0.30, $\hat{\beta}^*$ mostly between 0 and 0.4, no systematic difference across sectors (Figures 4 and 5): both features look universal
- Noise can fake roughness: a simulated $\alpha = 0$ observed with noise-to-signal ratio 0.3 at $\Delta = 0.1$ gives a mean OLS estimate of −0.47 but a mean NLLS estimate of 0.01 (Table 1, Panel C)
- 15-minute steps, $h = 20$: MSE ($\times 10^8$) 0.038 for log-HAR, 0.020 for RFSV, 0.019 for Power-BSS (Table 4, Panel A); daily steps, $h = 1$: RFSV is best, 0.493 ($\times 10^9$) (Table 5, Panel B): modelling persistence pays intraday and at long horizons

Appendix: Case Study, Intraday Application, Limits and Open Questions

- **Intraday application** (illustrative, not an exact replication) on the SPY 5-minute bars of the course data: $\hat{\sigma}_t^2$ = bipower variation of the 5-minute returns in each bin of $\Delta = 30$ or 65 minutes, divided by Δ and by the flexible Fourier seasonal factor Andersen & Bollerslev (1997)
 - ▶ OLS and NLLS $\hat{\alpha}$, Cauchy $\hat{\beta}^*$ and Gamma-BSS $\hat{\lambda}^*$ against Table 3, Panel A; log-HAR, RFSV and BSS forecasts, MSE, QLIKE and MCS against Table 5
 - ▶ Data limit: bars instead of every trade, so the pre-averaged estimator (Appendix A of the paper) is not feasible and the noise is larger; only 13 returns per 65-minute bin
- **Limits stated by the authors**
 - ▶ Realised measures are noisy estimates of integrated variance: the conclusions concern them, not the latent spot volatility (Sections 4.1, 4.2 and 6)
 - ▶ No rule for the NLLS bandwidth m (Sections 2.1 and 3.1); NLLS is too costly for rolling forecasts, which use the OLS $\hat{\alpha}$ (Section 5.1.1)
 - ▶ Forecasts assume a Gaussian X ; a non-Gaussian volatility of volatility is left out of scope (Section 5; footnote 3)
- **Open question:** why does long memory improve even short-horizon forecasts? The authors leave it to further research (Section 5.1.2)
 - ▶ Rolling $\hat{\alpha}$ (65 minutes, 2005–2014, median -0.36) is less rough in 2007–2008 and 2011 (Figure 3), and $\hat{\alpha}$ rises with log volume, slope 0.0091 (Figure 6): does Bitcoin (5-minute bars), a market that never closes, fit this pattern?

Appendix: The Discovery Loop, Reusable Prompts (1/2)

- Tools: an AI assistant (an LLM such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: [Semantic Scholar](#), [Elicit](#), [Google Scholar](#)
- 1. **Literature**: tabulate estimates by frequency, then check every DOI on [Crossref](#) (the title must match)
 - ▶ Tabulate published estimates of the jump share of equity-index variance by sampling frequency (daily, 5-minute, tick) and method; give the exact number, the table it comes from, and the DOI.
- 2. **Competing hypotheses**, with distinct testable implications
 - ▶ For three processes that give a daily excess kurtosis near 11 (Merton jumps, Heston volatility, drift bursts), derive the implications for returns standardised by ex-ante volatility, for bipower versus realised variance, and for the drift just before extreme moves.
- 3. **Data and code**: replicate the chapter's estimate first
 - ▶ Write Python that reads the S&P 500 daily closes of the course and fits the Merton model by maximum likelihood (Poisson mixture truncated at 12 jumps, several starting points). Target: $\hat{\lambda} = 102$ per year, jump share 71%

Appendix: The Discovery Loop, Reusable Prompts (2/2)

- 4. **Robustness**, pre-registered
 - ▶ Write a pre-registered plan: standardise returns by EWMA and by GARCH volatility known at $t-1$; split 1990-2007 and 2008-2026; test Heston (no jumps, the null) against Bates (SVJ); obtain LR critical values by parametric bootstrap under the Heston null.
- 5. **Critique**: ask the assistant to attack the inference
 - ▶ Act as a referee: why is the LR test of $\lambda = 0$ non-standard, what is the correct reference distribution, and could a Poisson mixture with frequent small jumps simply be a fat-tailed innovation?
- 6. **Report**: all frequencies and filters, bootstrap critical values, code and the AI log

Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Applied Computational Finance
- ☐  XFG Advanced Methods in Quantitative Finance

Code and data

- ☐ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_11



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